



CHITTAGONG UNIVERSITY OF ENGINEERING TECHNOLOGY  
Department of Electronics and Telecommunication Engineering

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VLSI Technology Sessional  
ETE 404

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Project Title : The complete SAP-1 with control sequencer .

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## Problem statement:

Using sap-1 computer calculate  $(5+2)-4=?$

## Program:

1:LDA 10

2:ADD 9

3:SUB 8

4:OUT

## Programming Steps:

- Turn on u\_mode pin.
- Turn on debug pin.
- Pulse the pc\_reset pin.

## LDA 10

- Set debug\_data to 0000 0000.
- Toggle mar\_in\_en and give a clock pulse.
- Turn off mar\_in\_en.
- Set debug\_data to 0001 1010.
- Toggle sram\_wr and give a clock pulse.
- Turn off sram\_wr and observe the data saved in address 0000.

### Store data(0000 0101) to memory location 1010:

- Set debug\_data to 0000 1010
- Toggle mar\_in\_en and give a clock pulse.
- Turn off mar\_in\_en.
- Set debug\_data to 0000 0101.
- Toggle sram\_wr and give a clock pulse.
- Turn off sram\_wr and observe the data saved in address 1010.

### ADD 9

- Set debug\_data to 0000 0001.
- Toggle mar\_in\_en and give a clock pulse.
- Turn off mar\_in\_en.
- Set debug\_data to 0010 1001.
- Toggle sram\_wr and give a clock pulse.
- Turn off sram\_wr and observe the data saved in address 0001.

### Store data(0000 0010) to memory location 1001:

- Set debug\_data to 0000 1001
- Toggle mar\_in\_en and give a clock pulse.
- Turn off mar\_in\_en.
- Set debug\_data to 0000 0010.
- Toggle sram\_wr and give a clock pulse.
- Turn off sram\_wr and observe the data saved in address 1001.

## SUB 8

- Set debug\_data to 0000 0010.
- Toggle mar\_in\_en and give a clock pulse.
- Turn off mar\_in\_en.
- Set debug\_data to 0011 1000.
- Toggle sram\_wr and give a clock pulse.
- Turn off sram\_wr and observe the data saved in address 0010.

### Store data(0000 0100) to memory location 1000:

- Set debug\_data to 0000 1000
- Toggle mar\_in\_en and give a clock pulse.
- Turn off mar\_in\_en.
- Set debug\_data to 0000 0100.
- Toggle sram\_wr and give a clock pulse.
- Turn off sram\_wr and observe the data saved in address 1000.

## OUT

- Set debug\_data to 0000 0011.
- Toggle mar\_in\_en and give a clock pulse.
- Turn off mar\_in\_en.
- Set debug\_data to 1110 0000.
- Toggle sram\_wr and give a clock pulse.
- Turn off sram\_wr and observe the data saved in address 0011.

- Turn off debug pin.
- Turn off u\_mode pin.

Now that programming is done, let us run our program.

## Fetch, Decode, Execute:

- Pulse the pc\_reset pin.
- Turn on o\_mode pin.

Give clock pulse and observe that the T-state change step by step and result update in reg\_A and reg\_B after execution each line of program .

## Timing Logic:

| T-state | LDA(0001)                    | ADD(0010)                    | SUB(0011)                    | OUT(1110)                |
|---------|------------------------------|------------------------------|------------------------------|--------------------------|
| T1      | Pc_out<br>Mar_in_en          | Pc_out<br>Mar_in_en          | Pc_out<br>Mar_in_en          | Pc_out<br>Mar_in_en      |
| T2      | Sram_rd<br>Ins_reg_in_en     | Sram_rd<br>Ins_reg_in_en     | Sram_rd<br>Ins_reg_in_en     | Sram_rd<br>Ins_reg_in_en |
| T3      | Pc_en                        | Pc_en                        | Pc_en                        | Pc_en                    |
| T4      | Ins_reg_out_en,<br>mar_in_en | Ins_reg_out_en,<br>mar_in_en | Ins_reg_out_en,<br>mar_in_en | Alu_out                  |
| T5      | Sram_rd,a_in                 | Sram_rd,B_in                 | Sram_rd,B_in                 | Don't care               |
| T6      | Don't care                   | Alu_out,A_in                 | Alu_out,A_in<br>Alu_sub      | Don't care               |